

MARÍA CASTELLÓ PONS

POSTDOCTORAL RESEARCH

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PROFESSIONAL PROFILE

Biomedical researcher focused on **glioblastoma biology, tumour plasticity and translational neuro-oncology**. PhD research combined orthotopic tumour models and patient-derived samples with **transcriptomic and proteomic analyses** to investigate treatment response and post-surgical remodelling of the tumour microenvironment. Additional experience in precision oncology, clinical data interpretation and international scientific project coordination.

CORE EXPERTISE

GLIOBLASTOMA · Tumour plasticity · EGFR-associated phenotypes · RT/TMZ response

MULTI-OMICS · Bulk RNA-seq · snRNA-seq · DIA proteomics · Spatial transcriptomics

BIOINFORMATICS · R · Python · Bioconductor · pandas · scikit-learn · Linux · Nextflow

EXPERIMENTAL MODELS · Orthotopic murine GBM · Stereotaxic surgery · Patient-derived tissue · IHC/IF

PROFESSIONAL EXPERIENCE

PhD Researcher — Neuro-Oncology

Instituto de Salud Carlos III (ISCIII), UFIEC · Madrid, Spain | Nov 2022 – 2026

- Investigated **glioblastoma (GBM) plasticity** and phenotype-dependent responses to **radiotherapy (RT)** and **temozolomide (TMZ)** using orthotopic murine models expressing EGFRwt or EGFRvIII.
- Integrated **bulk RNA-seq, single-nucleus RNA-seq and DIA proteomics** to characterise treatment-associated transcriptional and proteomic changes.
- Analysed paired **primary and recurrent patient-derived GBM samples** and investigated post-surgical tumour and microenvironment remodelling.
- Performed intracranial stereotaxic implantation, tumour monitoring, tissue collection and histological analysis; quantified IHC/IF images using **QuPath**.
- Contributed to **in-silico clinical trial design** for chemoradiotherapy optimisation with MOLAB/UCLM; performed data analysis and visualisation in **R and Python**.

Data Scientist | Project & Hospital Coordination

Iakan Health · Precision-oncology startup · Valencia, Spain | Oct 2021 – Nov 2022

- Coordinated scientific project activities and served as a liaison between **hospital teams, clinicians, healthcare professionals and technical colleagues** in precision-oncology projects.
- Supported collaboration with the **Hospital Ángeles de Puebla (Mexico)**, helping connect clinical needs with data-driven oncology solutions.
- Completed a **one-month on-site stay in Mexico** to support project coordination and collaboration with local healthcare stakeholders.
- Analysed and interpreted **clinical and genomic oncology datasets** to support personalised-treatment research and clinical decision-support activities.
- Represented Iakan Health internationally, including at **eMerge Americas Startup Showcase (Miami, USA, 2022)** and professional activities in Mexico.

Research Intern — Immunology & Autoimmune Diseases

GENYO – Pfizer-University of Granada-Junta de Andalucía Centre for Genomics and Oncology · Granada, Spain | Feb 2021 – Sep 2021

- Contributed to a **systemic lupus erythematosus** research project, supporting immunological assays and flow-cytometry analyses.

EDUCATION

PhD in Biomedical Sciences (Neuroscience) · UNED / Instituto de Salud Carlos III · 2022–2026. **Thesis completed; defence pending.**

MSc Regenerative Biomedicine · Universitat de València · 2020–2021.

BSc Biotechnology · Universitat de València · 2016–2020.

SELECTED PUBLICATIONS

• Castelló-Pons M†, Ramírez-González MÁ†, et al. VP3.15, a dual GSK-3β/PDE7 inhibitor, reduces glioblastoma tumor growth through changes in the tumor microenvironment in a PTEN wild-type context. *Neurotherapeutics*. 2025. doi:10.1016/j.neurot.2025.e00576. (†Equal contribution)

• Monago-Sánchez A, Mateos-Madrigal L, Carrión-Navarro J, et al., including Castelló-Pons M. Precision oncology in a dish: patient-derived glioma organoids to guide novel therapeutic strategies. *Biomarker Research*. 2026. doi:10.1186/s40364-026-00894-3.

• Perales-Patón M†, Italia M†, Castelló-Pons M†, et al. Optimizing chemoradiotherapy for malignant gliomas: a validated mathematical approach. Preprint. 2025. doi:10.48550/arXiv.2504.17481. (†Equal contribution)

SELECTED CONFERENCE CONTRIBUTIONS

• EANO Annual Meeting, Prague (2024) — Tumour phenotype determines radiotherapy response in glioblastoma.

• ASEICA 40th Anniversary Congress, A Coruña (2024) — Glioblastoma treatment optimisation and VP3.15 research.

• EACR Congress, Lisbon (2025) — Chemoradiotherapy optimisation using a validated mathematical model (co-author).

• ASCO Annual Meeting, Chicago (2023) — Breast-cancer prognosis tool using AI (Iakan Health contribution).

SELECTED FUNDED PROJECTS

• Mathematical Models for the Digital Transition in Neuro-Oncology (ISGBM) · EU Next Generation / ISCIII · 2022–2023 · €230,000.

• Characterisation of the Peri-Tumoral Area of Glioblastomas · ISCIII · 2022–2024 · €202,000.

• Iakan Project — AI for Improved Cancer Treatment · CREATEC-CV · 2020–2022 · €133,045.

TECHNICAL SKILLS

R, Python, Bioconductor, pandas, scikit-learn, Linux, Nextflow, bulk RNA-seq, snRNA-seq, DIA proteomics (Spectronaut v19.5), spatial transcriptomics (10x Visium), PCA/UMAP, orthotopic murine GBM models, stereotaxic surgery, patient-derived tissue, IHC/IF, QuPath, WB, PCR, cell culture, flow cytometry.

LANGUAGES & SELECTED TRAINING

Spanish and Valencian (native) · English (C1 Advanced, EF SET; Scientific English C1, ISCIII) · French (A2). Selected training: Wellcome Connecting Science (genomics datasets) · Johns Hopkins (Python for Genomic Data Science; Genomic Technologies) · ISCIII (Python, R, proteomics) · EACR/10x Genomics (spatial transcriptomics) · Laboratory animal work certification, Functions B & C.